

Guillermo Terrén-Serrano

Environmental Markets Lab (emLab), University of California Santa Barbara

4312 Bren Hall,
UC Santa Barbara
Santa Barbara, CA93106, USA

guillermoterren@ucsb.edu 
gterren.github.io 
gterren 
Google Scholar

I am a computational power systems researcher working on the planning and operation of low-carbon electricity systems under uncertainty using optimization and probabilistic forecasting methods. My research integrates power system modeling, machine learning, energy meteorology, and electricity market analysis to study grid reliability, resource adequacy, and infrastructure investment under evolving climate and socioeconomic conditions. As both a researcher and educator, my goal is to mentor the next generation of leaders in intellectually ambitious and inclusive research environments that cultivate scientific curiosity, technical excellence, and collaborative problem-solving.

Research Interests — *Power systems optimization; stochastic optimization; resource adequacy and reliability; probabilistic forecasting; electricity markets; low-carbon electricity systems; uncertainty quantification; machine learning for energy systems; energy meteorology and remote sensing.*

Professional Experience

Postdoctoral Scholar	2022–Present
Environmental Markets Lab (emLab) and Environmental Studies Program, University of California Santa Barbara, CA, USA	
CNSI Climate Innovation Postdoctoral Fellow	2023–2025
Environmental Markets Lab (emLab), University of California Santa Barbara, CA, USA.	
Postdoctoral Research Associate	2022–2023
Statistics and Applied Probability Department, University of California Santa Barbara, CA, USA.	
Graduate Research Assistant	2015–2022
Center for Emerging Energy Technologies (CEET) and Electrical & Computer Engineering Department, University of New Mexico, NM, USA.	
Project Engineer	2012–2013
Enerland Group, Zaragoza, Spain.	
Undergraduate Research Assistant	2012
Electronic Engineering Department, Institute of Technology for the Development (LACTEC), PR, Brazil.	

Education

Ph.D. in Electrical Engineering (Pass with Distinction)	2016–2022
“Intra-hour Solar Forecasting using Cloud Dynamics Features Extracted from Ground-Based Infrared Sky Images.” Department of Electrical & Computer Engineering, University of New Mexico, NM, USA	
<i>Advisor: Prof. Martínez-Ramón</i>	
M.Sc. in Power & Energy	2015–2016
“Machine Learning Approach for Global Solar Radiation Time-Series Forecasting.” Electrical & Computer Engineering Department, University of New Mexico, NM, USA.	
<i>Advisor: Prof. Martínez-Ramón</i>	

Peer-Reviewed Publications
([†]indicates equal contribution)

G. Terrén-Serrano[†], R. Deshmukh[†], S. Dukkipati, M. Meng, S. Prasad, A. Mileva and A. Gambhir. Cost-Optimal and Robust Decarbonization Pathways for India's Electricity System under Uncertainty. *Under review in Joule*, 2026.

G. Terrén-Serrano, R. Deshmukh and M. Martínez-Ramón. Probabilistic Day-Ahead Forecasting of System-Level Renewable Energy and Electricity Demand. *Nature Communications*, 2026.

G. Terrén-Serrano and M. Ludkovski. Extreme Day-Ahead Renewables Scenario Selection in Power Grid Operations. *Applied Energy*, 2025.

Q. Paletta, **G. Terrén-Serrano**, Y. Nie, B. Li, J. Bieker, W. Zhang, L. Dubus, S. Dev and C. Feng. Advances in Solar Forecasting: Computer Vision with Deep Learning. *Advances in Applied Energy*, 2023.

G. Terrén-Serrano and M. Martínez-Ramón. Deep Learning for Intra-Hour Solar Forecasting with Fusion of Features Extracted from Infrared Sky Images. *Information Fusion*, 2023.

G. Terrén-Serrano and M. Martínez-Ramón. Detection of Multiple Wind Velocity Fields using Ground-based Infrared Sky Images. *Knowledge-Based Systems*, 2023.

G. Terrén-Serrano and M. Martínez-Ramón. Kernel Learning for Intra-Hour Solar Forecasting with Infrared sky images and Cloud dynamic feature extraction. *Renewable and Sustainable Energy Reviews*, 2023.

G. Terrén-Serrano and M. Martínez-Ramón. Processing of Global Solar Irradiance and Ground-Based Infrared Sky Images for Solar Nowcasting and Intra-Hour Forecasting Applications. *Solar Energy*, 2023.

G. Terrén-Serrano and M. Martínez-Ramón. Geospatial Perspective Reprojections for Ground-Based Sky Imaging Systems. *IEEE Transactions on Geoscience and Remote Sensing*, 2022.

G. Terrén-Serrano and M. Martínez-Ramón. Comparative Analysis of Methods for Cloud Segmentation in Ground-Based Infrared Images. *Renewable Energy*, 2021.

G. Terrén-Serrano and M. Martínez-Ramón. Multi-Layer Wind Velocity Field Visualization in Infrared Images of Clouds for Solar Irradiance Forecasting. *Applied Energy*, 2021.

G. Terrén-Serrano, A. Bashir, T. Estrada and M. Martínez-Ramón. Girasol, a Sky Imaging and Global Solar Irradiance Dataset. *Data in Brief*, 2021.

A. Mammoli, **G. Terrén-Serrano**, A. Menicucci, T.P. Caudell, M. Martínez-Ramón. An Experimental Method to Merge Far-field Images from Multiple Long-Wave Infrared Sensors for Short-term Solar Forecasting. *Solar Energy*, 2020.

O. García-Hinde, **G. Terrén-Serrano**, M.Á. Hombrados-Herrera, V. Gómez-Verdejo, S. Jiménez-Fernández, C. Casanova-Mateo, J. Sanz-Justo, M. Martínez-Ramón and S. Salcedo-Sanz. Evaluation of Dimensionality Reduction Methods Applied to Numerical Weather Models for Solar Radiation Forecasting. *Engineering Applications of Artificial Intelligence*, 2018.

Manuscripts in Preparation

G. Terrén-Serrano, A. Sharma, R. Deshmukh, S. Prashad. Resource Adequacy in Electricity Systems under Demand Growth and Decarbonization. *In preparation*.

G. Terrén-Serrano and M. Ludkovski. Dynamic Intraday Probabilistic Forecasting Update of Renewable Power Generation. *In preparation*.

G. Terrén-Serrano and R. Deshmukh. Reliability Contribution of Renewable Energy Resources in Large Grids under Decarbonization. *In preparation*.

W. T. Wang, S. Prashad, **G. Terrén-Serrano** and R. Deshmukh. Employment, Health, and Environmental Effects of India's Power System Low-Carbon Transition. *In preparation*.

Conference Proceedings

G. Terrén-Serrano, R. Deshmukh, and M. Martínez-Ramón. Day-Ahead Operational Forecast of Aggregated Solar Generation Assimilating Mesoscale Meteorology Information. *IEEE PES Grid Edge Technologies Conference & Exposition*, San Diego, USA, 2025.

G. Terrén-Serrano and M. Martínez-Ramón. Explicit Basis Function Kernel Methods for Cloud Segmentation in Infrared Images. *Energy Reports*, 2022.

G. Terrén-Serrano and M. Martínez-Ramón. Wind Flow Estimation in Thermal Sky Images for Sun Occlusion Prediction. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022.

G. Terrén-Serrano and M. Martínez-Ramón. Segmentation Algorithms for Ground-Based Infrared Cloud Images. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022.

Grants and Funding

(*Indicates equivalent, postdoctoral scholars at UCSB are not eligible to serve as PI or co-PI)

Resource Adequacy and Long-Term Planning of India's Electricity System. *World Resource Institute (WRI)*, 2024–2026. Investigator. Total: \$450,000.

CNSI Climate Innovation Fund. *California Nano-System Institute*, 2023. Computing resources and conference travel. Co-PI*. Total: \$30,000.

Research Seed Program. *UCSB Institute of Energy Efficiency*, 2022. Software development and conference travel. Co-PI*. Total: \$50,000.

Stochastic Models, Indices & Optimization Algorithms for Pricing & Hedging Reliability Risks in Modern Power Grids. *ARPA-E PERFORM*, 2022–2023. Task Lead. Investigator. Total: \$3,500,000.

The New Mexico SMART Grid Center: Sustainable, Modular, Adaptive, Resilient, and Transactive. *NSF EPSCoR*, 2018–2025. Task Lead. Investigator. Total: \$20,000,000.

Honors and Awards

2023–2025 Climate Innovation Postdoctoral Fellowship. *California Nano-System Institute*. Competitive fellowship to research technologies for California's climate goal. Supported: extension of the joint day-ahead forecasting from solar generation (2022 Research Seed Program) to wind and electricity demand.

2015–2022 UNM-Avangrid Graduate Scholarship. *Avangrid (Iberdrola) Foundation.* Competitive scholarship to complete doctoral studies at the ECE department in the UNM under the King Felipe VI of Spain Endowed Chair.

2011 UZ-Santander Americampus Scholarship. *Santander Bank.* Competitive scholarship to support an academic exchange year at the PUC-PR, Curitiba, Brazil. Supported: senior project design.

Teaching and Mentoring Experience

Instructor of record:

ES 105 Renewable Energy Systems. Environmental Studies Program, UCSB.

Teaching assistant:

ECE 203 Circuit Analysis I. Electrical and Computer Engineering (ECE), UNM.

ECE 213 Circuit Analysis II. ECE, UNM.

ENG Introduction to Electronic Circuits. ECE, UNM.

ECE 314 Signals and Systems. ECE, UNM.

ECE 331 Data Structures and Algorithms. ECE, UNM.

Guest lecturer:

ECE 417 Advanced Machine Learning. Probabilistic Principal Component Analysis. ECE, UNM.

Mentor:

Steven De Jesus. EUREKA! Program Summer 2023, UCSB.

David Lopez. EUREKA! Program + CAMP Research School Year Internship 2022–2023, UCSB.

Alexander Santos-Lozada. CIESESE Program Summer 2017 + 2018, UNM.

Invited Talks and Conference Presentations

2025 Forecasting & Markets Workshop. *Presentation.* ESIG, TN, USA.

2025 Institute for Energy Efficiency (IEE). *Seminar.* UCSB, CA, USA.

2024 ARPA-E Energy Innovation Summit. *Exhibition Booth.* Department of Energy. TX, USA.

2024 Macro-Energy Systems (MES) Workshop. *Flash Talk + Poster.* Princeton University. NJ, USA.

2024 Energy Initiative. *Seminar.* Massachusetts Institute of Technology. MA, USA.

2023 International Energy Workshop (IEW). *Presentation.* Colorado School of Mines. CO, USA.

Datasets, Models, and Packages

G. Terrén-Serrano, R. Deshmukh, S. Dukkipati, M. Meng, S. Prasad, A. Mileva and A. Gambhir. *GridPath-India long-term (2020-2050) power system planning model data*. Dryad, (2026). Dataset & Model.

G. Terrén-Serrano, S. Prasad, and R. Deshmukh. *GridPath-India workshops: illustrative models for Rajasthan, Madhya Pradesh, and Tamil Nadu*. Zenodo, (2025). Model.

G. Terrén-Serrano, A. Bashir, T. Estrada, and M. Martínez-Ramón. *Girasol, a sky imaging and global solar irradiance dataset*. Dryad, (2021). Dataset.

Institutional Service and Community Involvement

Reviewer. Applied Energy; Renewable Energy; Knowledge-Based Systems; Solar Energy; Data in Brief; Wiley (book proposal).

Student Athlete. Rugby team, Universidad de Zaragoza, Zaragoza, Spain (2006–2011).

Spanish Translator. Parent-teacher conferences, South Valley Academy, NM, USA (2020).

Professional References

Prof. Ranjit Deshmukh (rdeshmukh@ucsb.edu). *Associate Professor*. Bren School of Environmental Science & Management, and Environmental Studies Program, UCSB, CA, USA.

Prof. Manel Martínez-Ramón (manel@unm.edu). *Professor*. Department of Electrical and Computer Engineering, UNM, NM, USA.

Prof. Michael Ludkovski (mludkov@ucsb.edu). *Professor*. Department of Statistics and Applied Probability, UCSB, CA, USA.

Dr. Andrea Mammoli (aamammo@sandia.gov). *Principal Member*. Sandia National Laboratories, NM, USA.